Aryan Vijay Bhosale

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Education

2025 - 2027 — **University of Maryland, College Park**, College Park, MD, USA
Master of Science in Computer Science
Advised by Prof. Dinesh Manocha & Prof. Ramani Duraiswami

2021 - 2025 — **Indian Institute of Technology, Bombay**, Mumbai, India
Bachelor of Technology with Honours in Mechanical Engineering
Minor in Artificial Intelligence and Data Science, Minor in Computer Science
Advised by Prof. Preethi Jyothi & Prof. Biplab Banerjee

Publications

- **Aryan Vijay Bhosale** et al. VIBE: Video Instruction-aligned Background music gEneration, **EMNLP (Findings) '26**
- **Aryan Vijay Bhosale** et al. DuplexWorld: Live Evaluation Benchmark for Speech-to-Speech (S2S) Voice Agents, **NeurIPS RTCA '26**
- **Aryan Vijay Bhosale** et al. TORUS: A Unified Benchmark for Unified Audio Models, **EMNLP SALMA '26**
- Lokegaonkar*, **Bhosale*** et al. Exploring Textual Guidance for Video-to-Music Generation Models, **NeurIPS (Creative AI Track) '26**
- Belham*, **Aryan Vijay Bhosale*** et al. Deep evolving semi-supervised anomaly detection
- **Aryan Vijay Bhosale** et al. Anomaly detection using Diffusion-based method *Equal contribution

Experience

Centific AI Research (CAIR) — Redmond, WA, USA
Research Intern: Multimodal AI Group, Audio & Speech Group — Jun '26 - Present

- Developed **DuplexWorld**, a live benchmark for speech-to-speech voice agents: GPT-Realtime, Gemini-Live, Grok Voice, Amazon Nova
- Developed **TORUS**, a unified benchmark testing generation-understanding self-coherence gap in unified audio models: NVIDIA Audex

Advisor: Abhishek Mukherji, Harshit Rajgarhia | Full-Duplex Speech, Speech Dialogue Models, Unified Audio Models, Live Evaluation Benchmarks

Dolby Laboratories — San Francisco, CA, USA
Research Collaborator: Dolby Advanced Technology Group (ATG) — Aug '25 - Present

- Developed and trained **VIBE** & **Video-Robin** series of **SOTA** text+video-to-music generation models in collaboration with **NVIDIA**
- Achieved **2.21x** inference speed-up, outperforming SOTA models on audio quality, diversity & alignment metrics across **3** benchmarks

Advisor: Lie Lu | Multimodal AI, Music Generation, Flow Matching, Diffusion Transformers, Distributed Training & Inference, PyTorch DDP

American Express — Gurugram, India
Data Science Intern: Model Interpretability (MI) Research — May '24 - Jul '24

- Pioneered an **ICE-based** framework to quantify confidence in MI techniques (**SHAP**, **PDP**, **ALE**) on production credit risk models
- Benchmarked interpretability techniques on **XGBoost** credit (**6M+ data**) & **Gen. Linear Model** attrition (**4M+ data**) risk models

Advisor: Aashish Prakash | Explainable AI, Statistical ML, XGBoost, statsmodels, SHAP, Scikit-Learn, Numpy

Brown University — Providence, RI, USA
Research Intern — May '23 - Aug '23

- Implemented a **conditional Diffusion Model** for generating optimal mechanical structures, **8x** better performance than SOTA **GAN**
- Integrated mechanically guided sampling methods & gradient-free optimizers (**Genetic Algorithms**, **CMAES**) using **PyTorch** & **evosax**

Advisor: Prof. Miguel Bessa | Generative AI, Deep Generative Models, Diffusion Models, DDPMs, PyTorch

Technical Skills

Programming Languages	Python | C++ | C | CUDA | Bash/Linux | LATEX | Java | SQL | MATLAB | R
Deep Learning	PyTorch | TensorFlow | Hugging Face (Diffusers, Transformers, PEFT, TRL) | JAX
Generative AI	Stable Diffusion & Stable Audio | LLaMa | LoRA | CLIP | VAE | LLaVa | Open AI, Claude API
Vision & Audio	OpenCV | Speechbrain | Fairseq | Torchaudio | Torchvision | Scikit-Image | YOLO | EasyOCR
ML Tools & Infrastructure	Git | Docker | Distributed Training (PyTorch DDP) | Multi-GPU Training | Hydra
Python Libraries	NumPy | Matplotlib | Scipy | Pandas | Scikit-Learn | wandb

Achievements

- Received sponsorship of **INR 4M+** from **NVIDIA** for building India's first self-driving car with custom drive-by-wire & **ROS** software
- Awarded **Undergraduate Research Award** for exceptional deep learning research on Diffusion Models for continual anomaly detection

Service

Program Committee, **NeurIPS Real-Time Conversational Agents** (RTCA) Workshop 2026 — *Sep '26 - Dec '26*
Reviewer, **Association for Computational Linguistics** (ACL) Rolling Review — *Aug '26 - Sep '26*
Reviewer, **Indian Conference on Computer Vision, Graphics and Image Processing** (ICVGIP) 2026 — *Sep '26 - Dec '26*
Reviewer, **ACM International Conference on Web Search and Data Mining** (WSDM) 2026 — *Sep '25 - Oct '25*
Research Assistant, Visual AI Lab, UK & Deep Visual Computing Lab, IIT Bombay — *Dec '23 - Aug '25*
Teaching Assistant, Computer Programming and Utilization Course (CS101), IIT Bombay — *Jan '24 - May '24*
Amazon Solve for Society Robotics Mentor — *Aug '23 - Sep '23*
Open Source Contributor, Hacktoberfest — *Oct '23 - Nov '23*
Academic Mentor, English Language Program Cabinet Member, Institute Web and Coding Convener, IIT Bombay — *Jul '24 - Dec '24*